Chemistry

Unit 2

Practice Exam 1

This sample exam paper has been prepared as part of the Pearson suite of resources for the Year 11, Unit 2, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes   Working time: 90 minutes

Materials required

An approved non-programmable calculator

Chemistry Data Sheet. This may be downloaded from the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	13	13	25	13	27
Section two: Short answer	7	7	30	36	33
Section three: Extended answer	3	3	35	42	40
				Total	**100**

Section one: **Multiple-choice** — 27% (13 marks)

This section has **13** questions. Answer **all** questions by circling the correct option.

Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Suggested working time: 25 minutes

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1 Which one of the following compounds would be expected to have the lowest vapour pressure at 25°C?

- **a** Water H_2O.
- **b** Octane (C_8H_{18})
- **c** Ethanol (C_2H_5OH)
- **d** Hydrogen chloride (HCl)

2 Which of the following gives the correct shape for each of the molecules listed?

	Linear	**V-shaped**	**Tetrahedral**
a	CO_2	H_2O	NH_3
b	H_2	CO_2	NH_3
c	HF	H_2S	CH_4
d	H_2O	NH_3	CH_4

3 Dispersion forces are the only intermolecular attraction between molecules of

- **a** HF.
- **b** OF_2.
- **c** NF_3.
- **d** CF_4.

4 The intermolecular forces that exist between molecules of HBr are

- **a** hydrogen bonding and dispersion forces.
- **b** hydrogen bonding and ion-dipole attractions.
- **c** dipole-dipole attractions and dispersion forces.
- **d** dipole-dipole attractions and hydrogen bonding.

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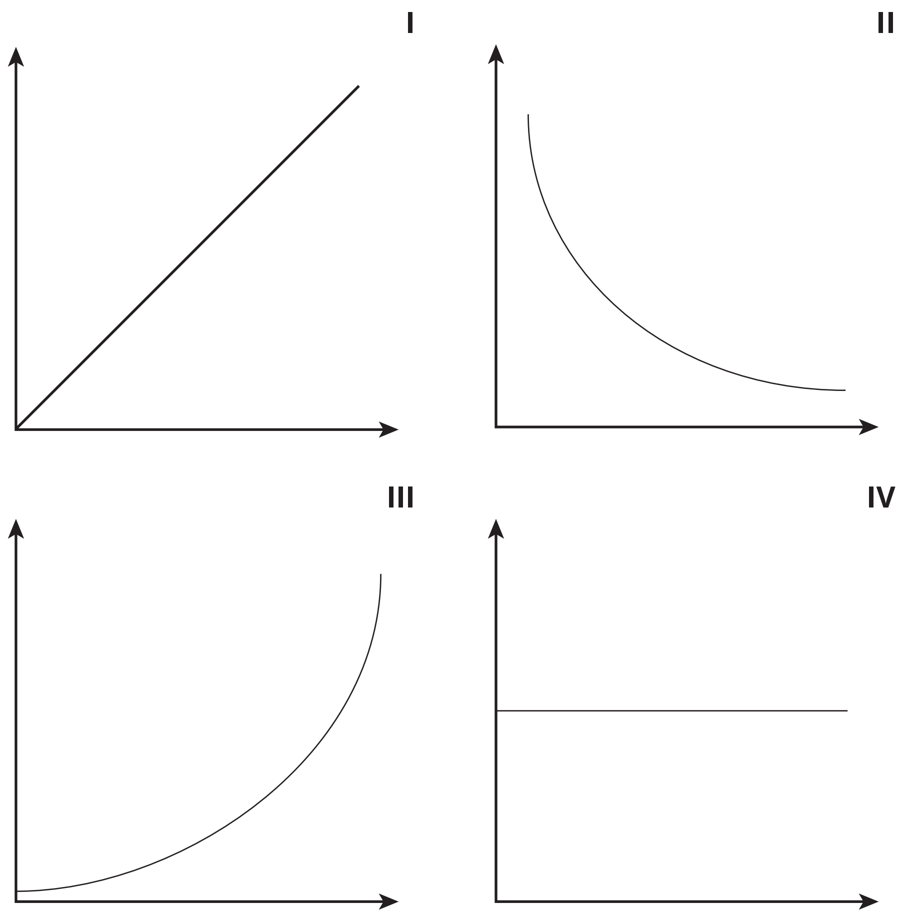
The following information refers to questions 5 to 8.

Consider the shapes of the following graphs:



5 For a given mass of a gas at a constant temperature, a graph of the relationship between the pressure (P) it exerts, and its volume (V) would have a shape similar to that of graph

a I.

b II.

c III.

d IV.

6 For a given mass of a gas at a constant pressure, a graph of the relationship between its temperature (T in K) and volume (V) would have a shape similar to that of graph

a I.

b II.

c III.

d IV.

7 A graph of the relationship between an amount of gas (n) and its volume (V) at constant temperature and pressure would have a shape similar to that of graph

a I.

b II.

c III.

d IV.

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8 For a given mass of a gas at constant temperature, a graph of the pressure it exerts (PV) and its volume (V) would have a shape similar to that of graph

a I.

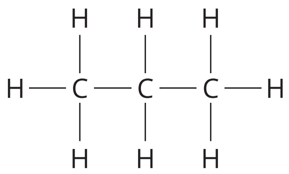
b II.

c III.

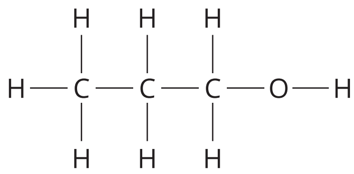
d IV.

9 Which one of the following compounds is least soluble in water?

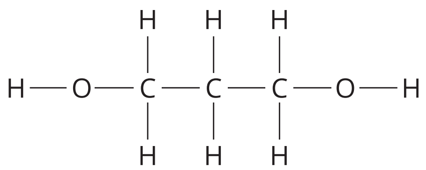
a



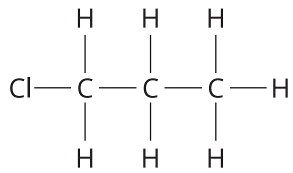
b



c



d



10 The pH range over which some indicators change colour is provided in the following table.

Indicator	pH range	Acid colour	Base colour
Methyl red	4.8–6.0	red	yellow
Bromothymol blue	6.0–7.6	yellow	blue
Phenolphthalein	8.2–10.0	colourless	red

Separate samples of a solution with unknown pH were tested with each indicator. The sample to which methyl red had been added turned yellow, the solution with bromothymol blue turned blue and the solution with phenolphthalein remained colourless.

The most accurate estimate of the pH of the solution is that the pH is between

a 4.8 and 7.6.

b 4.8 and 8.2.

c 6.0 and 10.0.

d 7.6 and 8.2.

11 The pH of 0.1 M HCN is 5, while the pH of 0.1 M HCl is 1. A concentrated solution of a weak acid is

a 5 M HCl.

b 5 M HCN.

c 0.1 M HCl.

d 0.1 M HCN.

12 The rate of a chemical reaction increases as the temperature increases. This is best explained by

a a decrease in the activation energy of the reaction.

b an increase in the activation energy of the reaction.

c an increase in the average kinetic energy of the particles.

d an increase in the number of collisions between particles.

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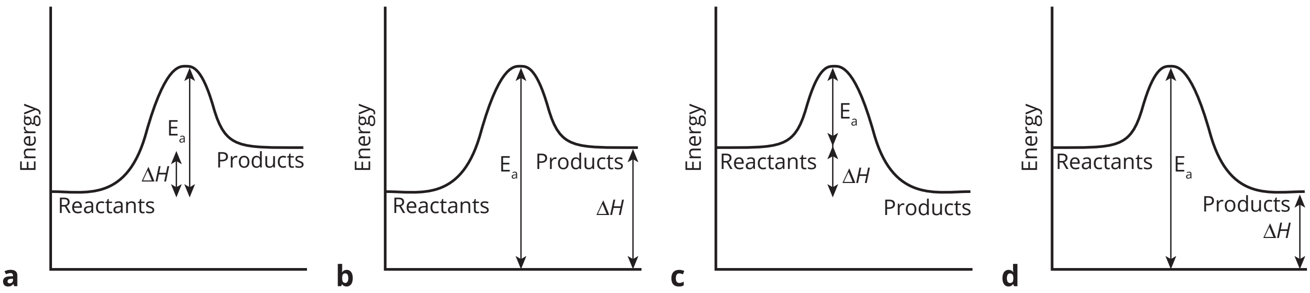
13 In each of the following energy profile diagrams ΔH represents the heat of reaction, and E_a the activation energy. Which one of the following diagrams correctly represents ΔH and E_a in an endothermic reaction?



End of section one

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Section two: Short answer 33% (36 marks)

This section has **8** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 30 minutes

Question 14 **(6 marks)**

The grid below includes names and formulas for 12 different substances.

Hydrogen sulfide (H_2S)	graphite (C)	Fluorine (F_2)	Carbon dioxide (CO_2)
Silicon dioxide (SiO_2)	Nitrogen (N_2)	Methane (CH_4)	Ammonia (NH_3)
Chloromethane (CH_3Cl)	Hydrogen fluoride (HF)	Oxygen (O_2)	Ethene (C_2H_4)

Identify substances from this grid that meet the following descriptions. Note that:

- the same substance may be used more than once
- a complete answer may require the inclusion of more than one substance.

Select a substance or substances

a with non-polar tetrahedral molecules.

b with polar tetrahedral molecules.

c with non-polar linear molecules.

d in which molecules are held to each other by dispersion forces only.

e that has/have intermolecular hydrogen bonds.

f that has/have molecules of the same shape as water molecules.

Question 15 **(6 marks)**

Bauxite is rich in alumina (Al_2O_3), which is used in the production of aluminium.

$2Al_2O_3(l) + 3C(s) \rightarrow 4Al(l) + 3CO_2(g)$

A particular sample of bauxite contains an average of 45% by mass of alumina (Al_2O_3).

a Calculate the maximum amount, in mol, of alumina that is present in 1.00 kg of bauxite. (2 marks)

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b Calculate the maximum mass, in gram, of aluminium that can be obtained from 1.00 kg of this bauxite. (2 marks)

c Calculate the volume, at STP, of carbon dioxide that would be produced from 1.00 kg of this bauxite. (2 marks)

Question 16 **(4 marks)**

When a 10.0 mL sample of a silver nitrate is added to excess sodium chloride solutions, 2.65 g of a precipitate forms.

a What is the name of the precipitate? ______ (1 mark)

b Write a balanced ionic equation for the formation of the precipitate. (1 mark)

c Calculate the concentration, in mol L^{-1}, of the silver nitrate solution. (2 marks)

Question 17 **(6 marks)**

a In 1883 Svante Arrhenius developed a theory that attempted to explain some of the properties of acids and bases. How did Arrhenius define an acid? (1 mark)

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b Give the name and molecular formula of one

i strong Arrhenius acid. ______ (1 mark)

ii weak Arrhenius acid. ______ (1 mark)

iii Write an ionic equation for the reaction between an Arrhenius acid and an Arrhenius base. (1 mark)

______

c Describe how the electrical conductivity differs between strong acids and weak acids. Explain this difference in terms of the Arrhenius theory. (2 marks)

______

______

______

______

______

______

______

Question 18 **(6 marks)**

Write ionic equations to represent each of the following reactions.

a Hydrochloric acid and magnesium powder. (2 marks)

______

b Nitric acid reacts with a solution of calcium hydroxide. (2 marks)

______

c Dilute sulfuric acid reacts completely with sodium carbonate solution. (2 marks)

______

Question 19 **(4 marks)**

In terms of collision theory, provide concise explanations for each of the following statements.

a When hydrochloric acid is added to some calcium carbonate and the reaction mixture is maintained at a constant temperature, the rate at which bubbles of carbon dioxide are observed decreases over time. (2 marks)

______

______

______

______

______

b Increasing the temperature of a reaction mixture significantly increases the rate of a reaction. (2 marks)

______

______

______

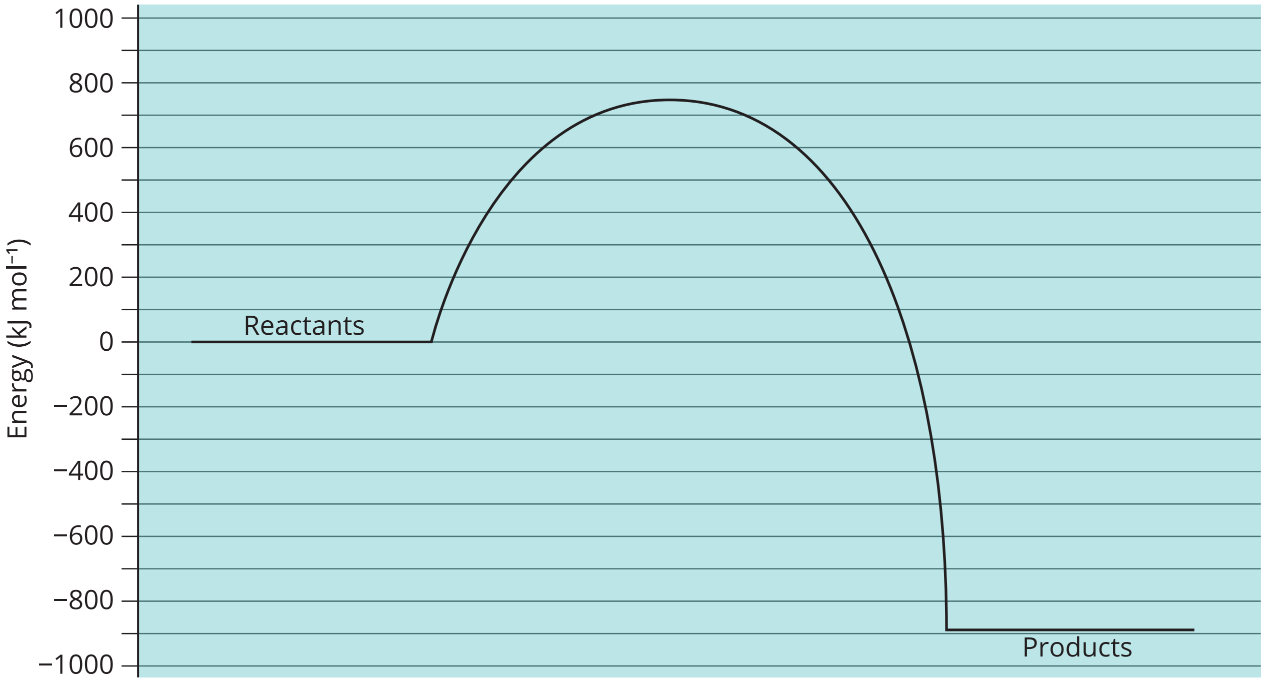
______

______

______

Question 20 **(5 marks)**

An energy profile diagram is provided for the combustion reaction of 1 mole of methane.



a Write a balanced equation of the combustion of methane. (1 mark)

____

b What is the value of Δ*H* for this reaction? ______________________ (2 marks)

c What is the value of the activation energy for this reaction? (2 marks)

____

End of section two

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Section three: **Extended answer** 40% (42 marks)

This section contains **3** questions. Answer **all** questions. Write your answers in the spaces provided.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.

Final answers to calculations should be expressed to the appropriate number of significant figures.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you use the spare pages to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

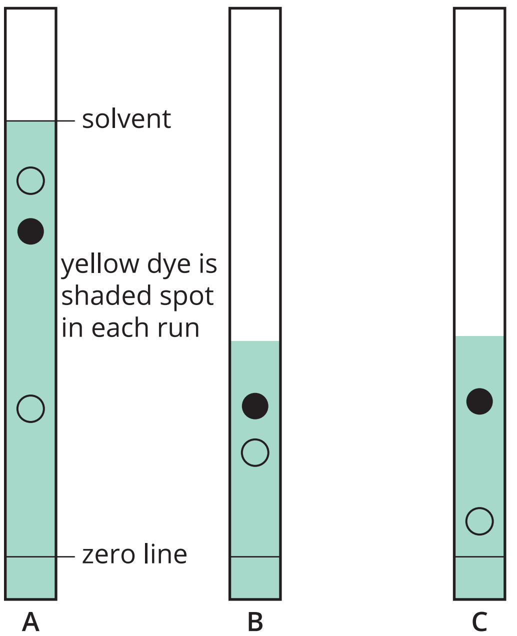
Suggested working time: 35 minutes

Question 21 **(10 marks)**

a There are several different chromatographic techniques including thin layer chromotography, gas chromotography and high performance liquid chromotography. They can all be used to determine the composition and purity of a substance.

What features are common to all types of chromatography? (2 marks)

b Food dyes extracted from a packet of sweets are tested using thin layer chromatography. Three different chromatograms are developed and shown in the following diagram. Ethanol is used as the solvent in A and B, and water in C.



i Explain why the components of the food dyes can be separated using thin layer chromatography. (2 marks)

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ii Is the yellow dye in chromatogram A likely to be the same as that in chromatogram B? Justify your answer. (1 mark)

iii Is the yellow dye in chromatogram B necessarily the same as in chromatogram C? Justify your answer. (1 mark)

iv A batch of food dye is suspected of being contaminated with an illegal colouring substance. Explain how you would determine, using thin layer chromatography, if the claim was correct. (2 marks)

c A mixture of three hydrocarbons, pentane (C_5H_{12}), octane (C_8H_{18}) and decane ($C_{10}H_{22}$), is passed through a gas chromatogram. The component that was eluted first identified as pentane, followed by octane and lastly decane.

i What can you deduce about the relative attraction of pentane to the packing of the column? (1 mark)

ii Which of the three hydrocarbons has the shortest retention time? (1 mark)

Question 22 **(21 marks)**

The solubilities of four different substances at 20°C and 80°C are given in the table below.

Substance	**Solubility in water (gL^{-1}) at**	
	20°C	**80°C**
Potassium sulfate	120	210
Sodium chloride	360	370
Graphite	Insoluble	Insoluble
Ethanol	Completely miscible	Completely miscible

a If 16 g of sodium chloride is mixed with 50 mL of water, what mass of sodium chloride will dissolve at 20°C? (1 mark)

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b If 9 g of potassium sulfate is mixed with 50 mL of water, what mass of potassium sulfate will remain undissolved at

i 20°C? ______ (2 marks)

ii 80°C? ______ (1 mark)

c If 1.5 g of graphite is mixed with 50 mL of water, what mass of graphite will dissolve at 20°C? (1 mark)

______

d What is the solubility of potassium sulfate at 20°C expressed in mole L^{-1}? (2 marks)

______

e You are given a mixture that contains 16 g of sodium chloride, 9 g of potassium sulfate and 15 g of graphite.

i Describe how you could obtain a pure sample of potassium sulfate and a pure sample of graphite from this mixture. (4 marks)

______

ii Use the data provided in the table above to explain why it would be difficult to obtain a pure sample of sodium chloride from the mixture. (2 marks)

______

f A small amount of potassium sulfate is dissolved in water.

i Write an ionic equation that represents the dissolving process when potassium sulfate is mixed with water. (1 mark)

______

ii Briefly describe the solution process in terms of the breaking of bonds and the making of new bonds as potassium sulfate is dissolved in water. (5 marks)

g Some ethanol is dissolved in water. Briefly describe the solution process in terms of the breaking of bonds and the making of new bonds as ethanol is dissolved in water. (3 marks)

Question 23 **(11 marks)**

You have been provided with three unlabelled bottles, A, B and C, each containing colourless solutions. The solutions are known to be barium chloride solution, potassium carbonate solution and ammonium chloride solution.

Solutions of each of the compounds were reacted with solutions of magnesium chloride, silver nitrate, sodium ethanoate, sodium hydroxide and sodium sulfate. The results of these reactions are summarised in the table below.

Solutions	**Bottle A**	**Bottle B**	**Bottle C**
magnesium chloride	no reaction	white precipitate	no reaction
silver nitrate	white precipitate	white precipitate	white precipitate
sodium ethanoate	no reaction	no reaction	no reaction
sodium hydroxide	white precipitate	no reaction	no reaction
sodium sulfate	white precipitate	no reaction	no reaction

a Use the information provided to identify the contents of bottles A, B and C. (3 marks)

Bottle A

Bottle B

Bottle C

b Suggest how a dilute solution of hydrochloric acid can be used to confirm the presence of potassium carbonate. Write a balanced ionic equation for any reaction that occurs.
(2 marks)

c Suggest how you would use a dilute solution of sodium hydroxide to test for the presence of ammonium chloride. Write a balanced ionic equation for any reaction that occurs.
(2 marks)

d Suggest one other test you could use to identify the contents of the three bottles. Describe how the results of this test will enable you to identify the three solutions. (4 marks)

End of questions

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